PUBLISHING_GUIDE.md

How to publish the a(3,n) = n+2 paper

1. Where things stand (as of this check)

Claim	Status
a(3,n) = n+2 for all n >= 4	Lower bound: explicit construction, checked by computer for n <= 120. Upper bound: computer-assisted potential certificate (443 states), verified by two independent implementations.
Lemma on horizontal extent and locality	Hand-proved, and checked by enumeration.
a(2,n) = 2 ceil(n/2)	Hand-proved. Probably elementary and possibly known.
Rectangular table (m <= 5)	Computed by CP-SAT, proved optimal by the solver. Not proved by hand.
Literature novelty	Not found in the sources searched (see Section 2). **Not established.**

Corrections made during this check:

- The earlier square-grid note said our lower bounds for n = 12–16 were not in OEIS. That was wrong. OEIS A271906 already gives a(12) >= 29, a(13) >= 32, a(14) >= 36 and a(15) >= 38. The note is corrected. Only n = 16 (value 40) is not listed.
- The earlier claim that the square-grid problem was “new” should not be repeated.

2. Literature check: what was searched and what it found

Searched: OEIS (A271906, A271907), arXiv full-text search, the Jánosik et al. survey (arXiv:2601.14465), the Károlyi–Solymosi preprint (arXiv:2607.22828), Shkredov–Solymosi (arXiv:2103.01645), PatternBoost (arXiv:2411.00566), Wu (arXiv:1605.00180), and web search.

Results:

- No source states a(3,n) or any result for a fixed number of rows.
- Square-grid results found: A271906 (values n <= 11, with lower bounds for 12–15), A271907 (no-isosceles variant), asymptotic bounds by Bloom and Pilatte, and the Károlyi–Solymosi lower bound.
- Related but different: Wu (arXiv:1605.00180) counts isosceles triangles on rectangular grids. It does not address the subset problem.
- A 2026 preprint (arXiv:2610.04194) uses AI-assisted search (sparse colouring) for isosceles-free grids. It concerns any-isosceles triangles, not right isosceles ones, but it is a precedent for disclosing AI assistance in this area.

Not done:

- zbMATH could not be queried successfully. Run a search there through your library, using "isosceles", "right triangle", "grid", "rows" and "subset", and check the reviews.
- Google Scholar and MathSciNet full-text searches.
- Read the full Jánosik et al. survey (only the first section was read) and the reference lists of the three 2026 papers.

Action: do these before submitting. If any source contains a(3,n) or an equivalent statement, the paper must be reframed or dropped.

3. Before you submit (required)

1. **Expert check.** Send the manuscript to two or three combinatorialists. Good sources: authors of the papers in Section 2 and colleagues in your department. Ask them specifically to (a) check the potential argument in Theorem 4 (the upper bound, with its telescoping step), and (b) say whether they know the result.
2. **Human-readable proof, if possible.** The upper bound currently depends on a finite certificate. Referees often ask for a human proof. If you can find a closed form for the potential, or a short argument, add it. If not, say clearly that the upper bound is computer-assisted (the manuscript already does).
3. **Authors.** The manuscript has a placeholder author line. Journals and arXiv require human authors who take responsibility for the content. An AI system cannot be an author. List the real authors and affiliations.
4. **AI disclosure.** Journals differ in their policies. Disclose the assistance in the manuscript (the "AI assistance" paragraph is a draft) and follow the target journal's wording.
5. **Reproducibility.** Upload the supplement folder (`code/`) to Zenodo or GitHub to get a DOI, and cite it in the paper. Its README lists the commands and the expected output. The certificate hash is in the paper.
6. **Title and abstract.** Keep "computer-assisted" in the abstract or title, so readers know the upper bound depends on a computer check.
7. **Clean up the paper.** Remove the date placeholder if you want a fixed date. Check the references: the Pilatte and Bloom citations are named in Károlyi–Solymosi and Jánosik et al., but a full bibliographic entry should be taken from the original source.

4. Where to post and submit

Step 1: arXiv. Post first to arXiv, primary category math.CO. Cross-list to cs.DM if you want. A first submission from a new account may need an endorsement, so check arXiv's current endorsement rules. arXiv gives a dated record, which helps if someone else publishes similar work.

Step 2: journal. Suitable venues, from best fit to more ambitious:

Journal	Why it fits	Notes
Electronic Journal of Combinatorics	Accepts short papers and computer-assisted results with code.	Free to read, well-regarded in combinatorics.
Journal of Integer Sequences	Good for the computed table and exact values.	Strong fit for a separate short note on the rectangular values.
Integers	Short notes in combinatorics and number theory.	Accepts computational results.
Discrete Mathematics	Broader readership.	More likely to ask for more depth.

Recommended plan: submit the main theorem (Sections 2–4 of the manuscript) to the Electronic Journal of Combinatorics or Discrete Mathematics. Submit the rectangular table and the two-row

result as a separate short note to the Journal of Integer Sequences, or add them to OEIS as new sequences.

Check the current journal policies yourself. I did not verify these venues' current rules on computer-assisted proofs, AI disclosure, or page limits.

5. Likely referee objections and responses

Objection	Response to prepare
"The upper bound is computer-assisted. Give a human proof."	The certificate is finite, and anyone can check it with the supplement. It was checked by two independent implementations. A closed form for the potential would be a stronger result, and we state this as open.
"This is known."	Describe the search in Section 2 precisely. If an expert knows a source, cite it and revise the novelty claim.
"The result is too small."	It is a complete determination of a(3,n) and a new method for narrow boards. It is a short note, and the paper should not claim more.
"Why not the square case?"	The square problem is open for n >= 12 (OEIS A271906). The method does not extend directly, since the state space grows with the number of rows.
"Is the lemma on horizontal extent correct?"	Yes. Lemma 1 has a short hand proof, and enumeration confirms it: 36 triangles fit in 3 rows, with maximum x-span 4 and maximum

6. Timeline (typical, not guaranteed)

- Week 1–2: literature check (Section 2) and expert feedback.
- Week 2–3: revise, add the human proof if found, post to arXiv.
- Months 1–6: journal review. Expect at least one round of revisions.

7. Files

- `paper/main.tex` and `main.pdf`: the manuscript.
- `code//`: code, certificate, audit scripts, README.
- `paper_note.pdf`: the earlier square-grid note, corrected.
- `rect_paper.pdf`: the rectangular-board note.